ORGANIC CUMULATIVE EXAMINATION

March 31, 1990

A recent seminar by Jencks reported on studies of the elimination reactions of β-Cyano Thioethers[1]. Some of the experimental results are summarized in Figures 1, 2, and 6 below. Interpret the data in these figures in terms of the scheme proposed by Fishbein and Jencks (Scheme I).

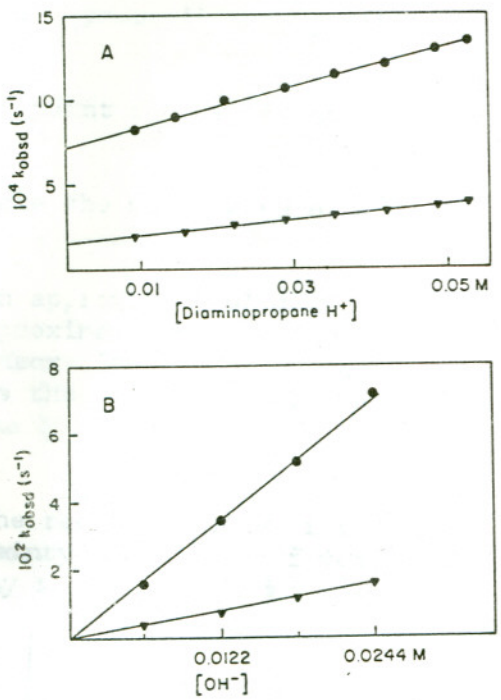


Figure 1. Dependence of k_{obsd} for elimination on catalyst concentration at 25 °C, μ = 1 M (KCl), 8.3% Me_2SO (v/v) in water. (A) Catalysis by 1,3-diaminopropane monocation (BH^{2+}/BH^+ = 5.7) of elimination from the β-protium (●) and deuterium (▼) pentafluorothiophenol adducts of fumaronitrile. (B) Catalysis by hydroxide ion of elimination from the β-protium (●) and deuterium (▼) *N*-methyl-2-thiopyridinium adducts of acrylonitrile.

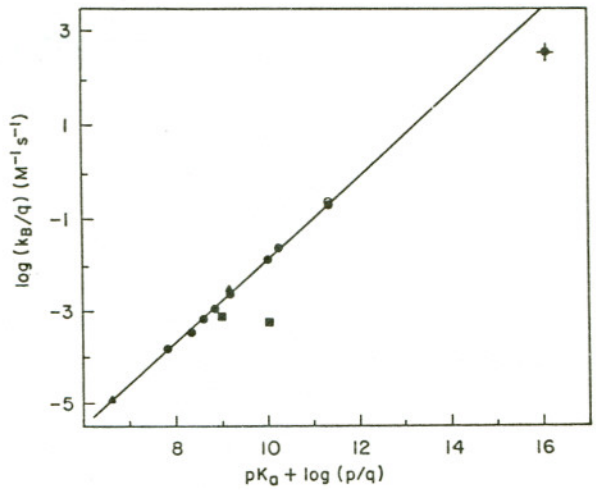


Figure 6. Statistically corrected Brønsted plot for catalysis of elimination from the pentafluorothiophenol adduct of fumaronitrile by different bases: (–●–), hydroxide ion; (●), primary amines; (▲), secondary amines; (■), hydrazine and ammonia. The solid line of slope 0.90 was calculated by least squares from the rate constants for primary amine catalysts.

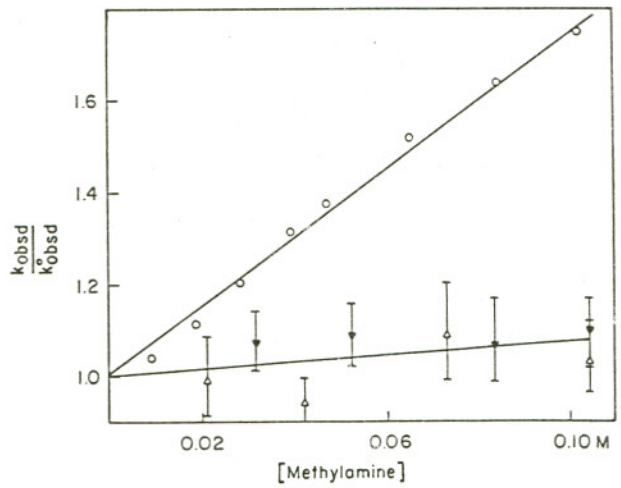
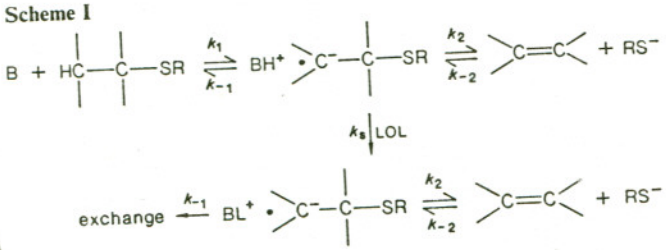


Figure 2. Comparison of catalysis by methylamine of elimination from the pentafluorothiophenol adduct of fumaronitrile in water at 25 °C (○) with catalysis by [$^2H_5$]methylamine of β proton exchange of the methanethiol adduct of fumaronitrile in D_2O at 23.8 °C in two different experiments (△ and ▼), in 8.3% Me_2SO (v/v), ionic strength 1 M (KCl). The value of k_{obsd} at zero buffer concentration, k_{obsd}^0, was obtained from the least-squares line for the points at all buffer concentrations. The buffer ratios were BL^+/B = 9.

Scheme I



[1] Fishbein, J.C.; Jencks, W.P. JACS 1988, 110, 5075. Elimination Reactions of β-Cyano Thioetheres: Evidence for a Carbanion Intermediate and Change in Rate-Limiting Step.